Test Task for the role of Intern/Junior IT Ops & Automation

Company name: Spendbase

Candidate's name : Oleg Malynyak

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Introduction

The goal of this work is to design a plan and technical assignment on an internal project. This will be a Physical Office Event Timer Clock device, that displays the elapsed time since a specific office event.
The clock must have Functional Requirements, such as Timer Display with show elapsed time with components as: “Years”, “Months”, “Days”, “Hours”, “Minutes”, “Seconds”. Also is required physical components, like button with opportunities to start/stop and reset Methods and somehow to input the initial event timestamp.
Device must have Real-time Operations, Power Management.

Idea

My core philosophy for this device was based on design simplicity, cost-effective components, and most importantly — given the technical requirements — the ability to operate under highly demanding conditions. The primary focus of the development was to ensure 24/7 functionality with minimal maintenance, efficient power management, and a reliable backup power source.
Additionally, the clock is equipped with Wi-Fi to utilise the NTP protocol for precise time synchronisation.

The development process began with the most critical element: the power supply.
First, there is the primary grid connection; under normal conditions, the clock will run directly from the main power. In the event of power outrage, the system will switch to a backup battery, which is kept continuously charged via solar panel.
As a third layer of protection, an independent RTC chip with own internal timer is integrated to ensure the time remains accurate even during prolonged internet outages.

Technical Specification

1. Hardware Components

Microcontroller: Wemos D1 Mini (based on ESP8266) with integrated Wi-Fi

Reason for selection: Chosen for its built-in Wi-Fi capabilities for NTP time synchronisation at a fraction of the cost of other controllers, compact form factor. Includes a built-in USB-to-Serial converter, making it easy to program and power without extra hardware.

Display: 0.91-inch OLED (SSD1306)

Reason for selection: This is one of the most cost-effective digital displays on the market. Using a simple 4-wire I2C interface.

Real-Time Clock: DS3231 Precision RTC

Reason for selection: Selected to provide a reliable hardware backup for timekeeping. This chip ensure the device maintains accurate time, preventing the need for manual recalibration.

Charging Controller: TP4056 (Module)

Reason for selection: An industry-standard, low-cost solution for lithium battery management. It combines charging and protection circuits into a single, inexpensive component, ensuring battery safety.

Voltage Regulator: MT3608 Boost Converter

Reason for selection: Chosen for its affordability and high availability. It allows the use of a single-cell Li-ion battery to power 5V components, making the overall power system significantly cheaper than using multi-cell battery packs or complex buck-boost regulators.

Diodes: 1N5819 Schottky

Reason for selection: Widely available and inexpensive. Provides a simple “passive” way to handle dual-power inputs(USB/Solar) without need for costly active circuits.

Backup Battery: CR2032 Lithium Coin Cell

Reason for selection: Acts as the "final line of defence" for timekeeping. It ensures the DS323 RTC chip maintains the correct time for several years, even if the main Li-ion battery is completely depleted.

Primary Power Source: 5V USB

Reason for selection: Acts as the main, stable power supply for consistent 24/7 operation in an office environment.

Secondary Power Source: 5V Solar Panel (Power Buffer Strategy)

Reason for selection: Serves as a sustainable backup energy harvester. It provides power autonomy during potential grid outages and continuously trickle-charges the battery buffer to ensure long-term stability.

User Input: Tactile push Button (Momentary)

Reason for selection: Simple, robust mechanical solution for the "Start/ Reset" function.

I2C Pull-up Resistors:

Required to maintain the SDA and SCL signal lines at a stable logic "HIGH" state. This ensures reliable data communication between the MCU, the OLED display, and the RTC module by preventing floating signals.

Current Programming Resistor: R_PROG

Reason for selection: A high-precision resistor connected to the TP4056 PROG pin. It is essential for setting the maximum charging current to 1A, ensuring a safe and efficient charging rate for the Li-ion battery.

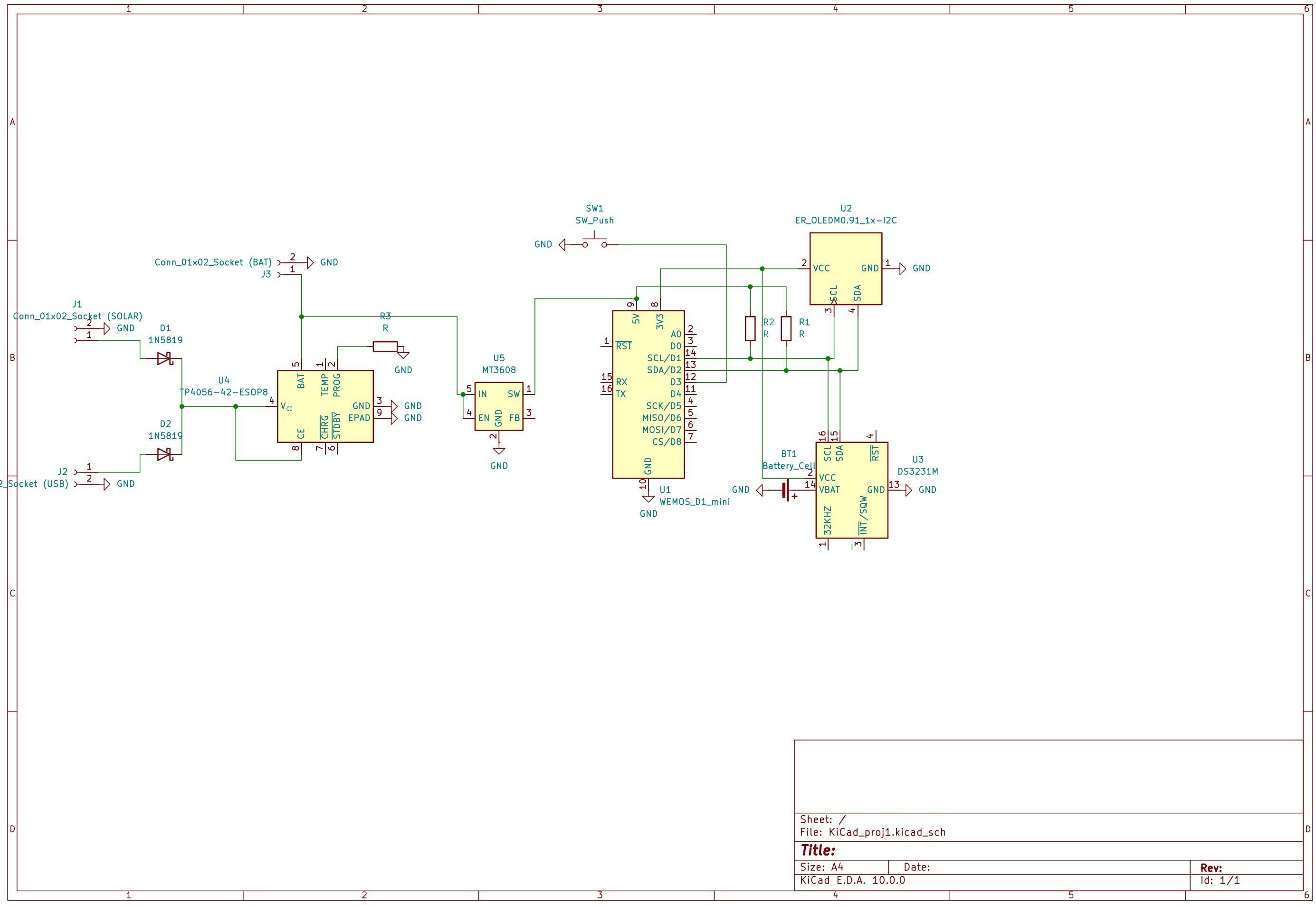


Figure 1: Complete System Schematic Diagram.

2. Software Architecture

Core Concept:

Clock tracks how much time has passed since a specific moment. The system is designed to be reliable, simple, and easy to use via three interfaces:

- Hardware: A physical button to reset the timer.
- Visual: An OLED screen for real-time monitoring.
- Digital: A Web interface to set custom dates and restart the timer via Wi-Fi.

Development Tools:

The software was built using:
- VS Code + Arduino Extension: The primary editor for writing code and managing the project.
- Arduino IDE 2.3+: Used for managing libraries.
- Git & GitHub: Used for version control.

Software Provenance (Origin):

- Open Source Base: The code is not written from scratch. I used open-source logic for the Web Server and RTC from GitHub (specifically based on the TimerRelay project by *ajithvasudevan*).
- Refactoring: The original code was heavily simplified. I removed unnecessary things to keep it “clean."
- AI (LLM) Integration: I used Large Language Models (LLM) to integrate the OLED display and optimize the EEPROM memory storage.

Data Flow:

How information moves through the system:

- Input: The DS3231 RTC module provides the current time via the I2C bus.
- Logic: The ESP8266 calculates the difference: Current Time - Saved Event Time.
- Storage: The Event Time is saved in the EEPROM (permanent memory), so the timer does not reset if the power goes out.

Output:

- OLED: Displays days, hours, minutes, and seconds. It updates every loop. Approximately every 2-10 milliseconds depending on if we are using web in the moment.
- Web Server: Sends the status to your phone's browser when you connect to the "My_Event_Timer" Wi-Fi.

Dependencies (Libraries):

I used libraries from Adafruit to handle the hardware:

- RTClib: To talk to the clock module.
- Adafruit SSD1306 & GFX: To draw text and numbers on the screen.
- ESP8266WebServer: To create the Wi-Fi hotspot and the settings page.

Operational Modes:

- Standalone Mode: The device works on its own. You can see the time on the screen and reset it with the physical button.
- Interactive Mode: You can connect via Wi-Fi to a special webpage (192.168.4.1). Here, you can use a Date Picker to choose a specific date from the past.

Design

Design Concept:

The enclosure is designed as a functional geometric prism (pyramid-style). This shape serves a dual purpose:

- Ergonomics: The perfect viewing angle for the OLED display when placed on a desk.
- Solar Efficiency: The rear face is also angled, allowing the solar panel to face upwards and backwards to capture maximum light.

Parametric Modeling (OpenSCAD):

The case was built using OpenSCAD, a code-based modeling tool.

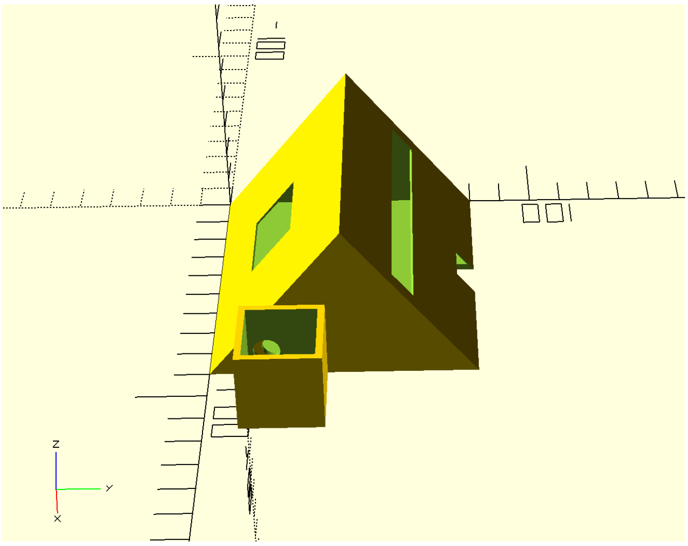
- The Power of Code: The entire design is defined by variables (parameters).
- Easy Upgrades: If the hardware changes (e.g., a larger solar panel or a different screen), we don't need to redraw the model. We simply update a few lines of code.

Physical Dimensions:

- Main Body: 90×80×50 mm.
- Button Module: A dedicated 25×25×25 mm cube attached to the side for the reset button.
- Cutouts: Custom-fit openings for the 0.96" OLED display, a 80×45 mm solar panel, and a rear Micro-USB access port.

Manufacturing:

- The design is optimized for 3D printing (FDM).



Bill of Materials

Category	Component	Description	Qty	Unit Price (Est.)	Total
Control	Wemos D1 Mini	ESP8266 Wi-Fi Microcontroller	1	$4.50	$4.50
Time	DS3231 RTC	High-precision Real-Time Clock module	1	$3.00	$3.00
Time	CR2032 Battery	Backup battery for RTC timekeeping	1	$0.50	$0.50
Display	OLED 0.96"	SSD1306 I2C Digital Display	1	$3.50	$3.50
Power	Li-ion Battery	18650 Rechargeable Battery (Buffer)	1	$5.00	$5.00
Power	TP4056 (U4)	Li-ion Battery Charging Controller	1	$0.80	$0.80
Power	MT3608	DC-DC Boost Converter (Step-up to 5V)	1	$1.20	$1.20
Power	Solar Panel	5V Solar Energy Harvester	1	$2.50	$2.50
Components	I2C Resistors	4.7k Ohm Pull-up resistors (SDA/SCL)	2	$0.10	$0.20
Components	R_PROG Resistor	High-precision resistor for TP4056 current	1	$0.10	$0.10
Components	Diodes 1N5819	Schottky diodes for power switching	2	$0.20	$0.40
Misc	Button & Wires	Tactile button and connection wires	1	$1.00	$1.00
Enclosure	3D Case	PLA/PETG Enclosure (OpenSCAD design)	1	$2.00	$2.00
TOTAL					$24.70